

1. Business Problem Statement

The primary objective of this project is to transform raw retail data from multiple sources (CSV files) into an actionable analytical ecosystem. The business faced challenges in tracking performance across different years (2017–2018), understanding customer demographics, and identifying product quality issues through returns. By building a structured SQL database, we aim to provide a "Single Source of Truth" that enables stakeholders to make data-driven decisions regarding sales growth, inventory efficiency, and targeted marketing.

2. Key Business Questions

To address the business problem, we developed SQL queries to answer the following critical questions:

- **Financial Growth:** What is our annual revenue and profit trend, and what is the running total of revenue across 2017 and 2018?
- **Brand Performance:** Which are the top 5 most profitable brands in our inventory?
- **Operational Efficiency:** Which stores are most efficient based on "Sales per Grocery Square Foot"?
- **Customer Profiling:** How is sales volume distributed across different gender groups and yearly income brackets?
- **Market Expansion:** Which countries generate the highest sales volume to support future expansion decisions?
- **Product Quality:** Which products have the highest return rates, and how do "Low Fat" products perform compared to regular ones?
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3. Data Modeling: The Star Schema

We organized the database using a **Star Schema** to optimize query performance and ensure ease of use for Power BI reporting. This involves separating data into **Facts** (quantitative measurements) and **Dimensions** (descriptive attributes).

Relationship Mapping (The Plan):

To enable seamless filtering, we established the following relationships:

- **Sales Fact Table:** Connects to **Customers** via `customer_id`, to **Products** via `product_id`, to **Stores** via `store_id`, and to the **Calendar** via `transaction_date`.
- **Returns Fact Table:** Connects to **Products** via `product_id`, to **Stores** via `store_id`, and to the **Calendar** via `return_date`.

Table Name	Type	Key Connectivity
Sales	Fact	customer_id, product_id, store_id, date
Returns	Fact	product_id, store_id, date
Products	Dimension	Links to Sales and Returns via product_id
Customers	Dimension	Links to Sales via customer_id
Stores	Dimension	Links to Sales and Returns via store_id
Calendar	Dimension	Links to Sales and Returns via date

4. Data Preparation & Optimization

We didn't just load the data; we cleaned and transformed it through **SQL Views** to make it "Power BI Ready." Key actions included:

- **Fact Unification:** Combined `Sales_2017` and `Sales_2018` into a single view (`vw_AllSalesFact`) using `UNION ALL` to allow for a continuous timeline analysis.
- **Denormalization:** Merged the `Region` and `Stores` tables into `vw_CleanStores`. By using a **LEFT JOIN**, we ensured **Data Integrity**, keeping all store records even if regional data was missing.
- **Data Cleaning:**
 - In `vw_CleanCustomers`, we combined names, calculated `current_age`, and handled `NULL` values.

- In vw_CleanProducts, we calculated unit_profit at the SQL level to reduce the processing load on Power BI.
- In vw_CleanStores: We calculated a new metric, display_area_pct, to determine the percentage of the store dedicated to groceries. This allows the business to analyze space utilization and efficiency across different store types and regions.
- In vw_CleanCalendar: we transformed the raw date table into a comprehensive dimension by extracting specific time attributes, such as year, month_name, quarter, and day_name. This was done to provide intuitive navigation for end-users, allowing them to drill down from yearly trends to specific weekdays with ease.